Chemistry for Computer Science

Course 26CYB1002J · Lecture 2, Tutorial 1, Practical 2, Credits 4

Made by Ajlikeadj

Contents

Unit 1: Quantum Chemistry

- Classical to quantum mechanics: postulates
- The Schrödinger equation: derivation and meaning
- Particle in a 1-D box: full solution
- The hydrogen atom: full solution
- Molecular orbitals and Hückel theory
- In the lab

Unit 2: Periodic Properties & Cheminformatics

- Effective nuclear charge and Slater's rules
- Ionization energy, electron affinity, electronegativity, polarizability
- Tutorial: Zeff problems with solutions
- Practical: hardness of water by EDTA titration
- Basics of cheminformatics
- Representing molecules in two and three dimensions
- The SMILES format
- Standard software tools
- Chemical databases and how to use them
- Practical: AI classification model for drug toxicity

Unit 3: Thermodynamics & Electrochemistry

- Basic terms and the four laws
- Thermodynamic functions: internal energy and enthalpy

- Entropy: meaning and estimation
- Gibbs free energy: derivation and use
- Case study: ammonia synthesis (Haber–Bosch)
- Free energy and EMF of cells
- The Nernst equation and electrode potentials
- Potentiometric titration of Fe^{2+} with $K_2Cr_2O_7$

Unit 4: Materials Chemistry

- Magnetic materials
- Superconducting materials
- Band structure of solids and the role of doping
- Semiconductor materials
- Introduction to nanomaterials
- Properties of nanomaterials
- Types of nanomaterials
- Quantum dots
- Applications of nanomaterials
- Practical: strength of an acid with a pH meter
- Practical: band gap of a semiconductor
- Practical: HCl and acetic acid mixture by conductometry

Unit 5: Spectroscopic Analysis

- Electromagnetic radiation
- Vibrational and electronic transitions
- The Born–Oppenheimer approximation
- UV–visible spectroscopy
- Infrared spectroscopy
- Band gap from UV–visible spectra (Tauc plot)
- Nuclear magnetic resonance: introduction
- NMR explained: chemical shift, integration and splitting
- Applications of NMR, with a worked problem
- Practical: Hooke's-law IR spectra simulation
- Practical: excitation energy of an organic molecule
- Viva: questions and answers

Unit 1: Quantum Chemistry

Wavefunctions, the Schrödinger equation, the particle in a box, the hydrogen atom and molecular orbitals.

Classical to quantum mechanics: postulates

S1 · Lecture

Classical (Newtonian) mechanics assigns a particle an exact position and momentum at every instant and predicts its path. It works for cricket balls and planets, but around 1900 three experiments showed it fails for electrons and light.

Three failures of classical physics

1. **Blackbody radiation.** Classical theory predicted infinite energy at short wavelengths (the "ultraviolet catastrophe"). Planck fixed it by assuming that oscillators exchange energy only in packets:

   ```
   E = hν      h = 6.626×10⁻³⁴ J s
   ```

2. **Photoelectric effect.** Light ejects electrons only if its frequency exceeds a threshold, however bright it is. Einstein: light is made of photons of energy hν, and one photon frees one electron:

   ```
   KE(max) = hν − φ      (φ = work function)
   ```

3. **Atomic spectra.** Hydrogen emits only certain wavelengths (lines), not a continuum. The lines obey the Rydberg formula, and classically an orbiting electron should spiral into the nucleus:

   ```
   1/λ = R_H (1/n₁² − 1/n₂²)      R_H = 1.097×10⁷ m⁻¹
   ```

Matter waves

de Broglie proposed that every particle has a wavelength, $\lambda = h/p$. Example: an electron accelerated through 100 V has $p = \sqrt{(2meV)}$, so $\lambda = h/\sqrt{(2meV)} \approx 0.123$ nm, comparable to atomic spacing (this is why electrons diffract from crystals). A 0.15 kg ball at 40 m/s has $\lambda \approx 10^{-34}$ m, far too small to notice.

The postulates of quantum mechanics

1. **Postulate 1: state.** The state of a system is completely described by a wavefunction $\psi(x,y,z,t)$.

2. **Postulate 2: Born interpretation.** $|\psi|^2 d\tau$ is the probability of finding the particle in volume $d\tau$. Since the particle must be somewhere, $\int |\psi|^2 d\tau = 1$ (normalization).

3. **Postulate 3: operators.** Each observable has an operator: position $\hat{x} = x$, momentum $\hat{p}_x = -i\hbar\, d/dx$, energy $H = -(\hbar^2/2m)\nabla^2 + V$.

4. **Postulate 4: measurement.** A measurement can only give an eigenvalue a of the operator, where $\hat{A}\psi = a\psi$. Check: $\hat{p}$ acting on e^(ikx) gives $-i\hbar(ik)e^{ikx} = \hbar k\, e^{ikx}$, so the momentum is $\hbar k = h/\lambda$, matching de Broglie.

5. **Postulate 5: dynamics.** The state evolves by $i\hbar\, \partial\Psi/\partial t = H\Psi$.

Uncertainty principle

Because $\hat{x}$ and $\hat{p}$ do not commute, position and momentum cannot both be known exactly:

$$\Delta x \cdot \Delta p \geq \hbar/2$$

It is not instrument error; it is built into wave behaviour. Confining an electron to a small region (small Δx) forces a large Δp, which is the reason a confined particle has zero-point energy.

The Schrödinger equation: derivation and meaning

S2 · Lecture

Schrödinger did not prove his equation from earlier physics; it is a postulate that fits experiment. But it can be motivated in four short steps, by combining the wave picture of the electron with energy conservation.

1. **Start with a standing wave.** A wave in one dimension with wavelength λ can be written $\psi = A \sin(2\pi x/\lambda)$. Differentiate twice:

$$\frac{d\psi}{dx} = \frac{2\pi}{\lambda} A \cos\left(\frac{2\pi x}{\lambda}\right)$$
$$\frac{d^2\psi}{dx^2} = -\left(\frac{2\pi}{\lambda}\right)^2 A \sin\left(\frac{2\pi x}{\lambda}\right) = -\left(\frac{2\pi}{\lambda}\right)^2 \psi$$

2. **Bring in the electron's wavelength.** de Broglie: $\lambda = h/p$, so $2\pi/\lambda = 2\pi p/h = p/\hbar$ (where $\hbar = h/2\pi$). Substituting:

$$\frac{d^2\psi}{dx^2} = -\frac{p^2}{\hbar^2} \psi$$

3. **Bring in energy.** Total energy = kinetic + potential: $E = p^2/2m + V(x)$. Rearranged, $p^2 = 2m(E - V)$. Substituting this for p^2:

$$\frac{d^2\psi}{dx^2} = -\frac{2m}{\hbar^2}(E - V) \psi$$

4. **Rearrange.** Multiply by $-\hbar^2/2m$ and move $V\psi$ across:This is the time-independent Schrödinger equation in one dimension.

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x) \psi = E \psi$$

Reading the equation

The first term is the kinetic-energy operator. $d^2\psi/dx^2$ measures how sharply ψ curves, so a more tightly curved (more "wiggly") wavefunction has higher kinetic energy. The second term is potential energy. Together they form the Hamiltonian operator H, giving the compact form $H\psi = E\psi$: an operator acting on a function returns the same function times a number. Such a function is an eigenfunction and the number E is its eigenvalue, the allowed energy. In three dimensions d^2/dx^2 becomes the Laplacian $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$.

Why energy becomes quantized

Mathematically the equation has solutions for every E. Physically ψ must be (1) single-valued, (2) continuous with a continuous slope where V is finite, (3) finite everywhere, and (4) normalizable: $\int |\psi|^2 d\tau = 1$. Only certain values of E allow a solution that meets these boundary conditions, and that is the origin of quantization. The next topic shows this happening.

Time-dependent form

When the state changes in time: $i\hbar\,\partial\Psi/\partial t = H\Psi$. If V does not depend on time, $\Psi(x,t) = \psi(x)\cdot e^{(-iEt/\hbar)}$, and the spatial part obeys the equation above. These stationary states have $|\Psi|^2$ independent of time.

$$\hat{H}\psi = E\psi \qquad \hat{H} = -(\hbar^2/2m)\nabla^2 + V \qquad |\psi|^2 = \text{probability density}$$

Particle in a 1-D box: full solution

S3, S9–S10 · Tutorial & Practice

The model. A particle of mass m moves along x between impenetrable walls at x = 0 and x = L. Inside, the potential is zero; outside, it is infinite.

$$V(x) = 0 \quad \text{for } 0 < x < L$$
$$V(x) = \infty \quad \text{for } x \le 0 \text{ or } x \ge L$$

Since the particle cannot be where V is infinite, $\psi = 0$ outside the box. Only the inside needs solving.

1. **Write the equation inside the box (V = 0).**

$$-(\hbar^2/2m)\,\psi'' = E\,\psi \quad \rightarrow \quad \psi'' = -k^2\,\psi, \quad \text{where } k^2 = 2mE/\hbar^2$$

2. **Solve it.** Functions whose second derivative is $-k^2$ times themselves are sine and cosine, so

$$\psi(x) = A\sin(kx) + B\cos(kx)$$

3. **Apply the boundary condition at x = 0.** ψ must be continuous with the outside value 0:

$$\psi(0) = A\cdot 0 + B\cdot 1 = 0 \quad \rightarrow \quad B = 0 \qquad \text{so } \psi = A\sin(kx)$$

4. **Apply the boundary condition at x = L.** A = 0 would give $\psi = 0$ everywhere (no particle), so sin(kL) = 0, hencen = 0 is excluded ($\psi = 0$); negative n only flips the sign of ψ, the same state. Quantization has appeared from the boundary conditions alone.

$$\psi(L) = A\sin(kL) = 0$$
$$kL = n\pi \quad \rightarrow \quad k = n\pi/L, \quad n = 1, 2, 3, \ldots$$

5. **Find the energies.** From $k^2 = 2mE/\hbar^2$ we have $E = \hbar^2k^2/2m$. Insert $k = n\pi/L$ and $\hbar = h/2\pi$:

$$E_n = \hbar^2n^2\pi^2/(2mL^2) = n^2h^2/(8mL^2)$$

6. **Normalize.** The particle is somewhere in the box, so the total probability is 1:(the integral of $\sin^2$ over a whole number of half-waves is half the length.)

$$\int_0^L A^2\sin^2(n\pi x/L)\,dx = A^2\cdot(L/2) = 1 \quad \rightarrow \quad A = \sqrt{(2/L)}$$
$$\psi_n(x) = \sqrt{(2/L)}\cdot\sin(n\pi x/L)$$

What the result tells us

Energy is quantized: only n^2 multiples of $h^2/8mL^2$. Zero-point energy: the lowest level is n = 1, not 0, so $E_1 = h^2/8mL^2 > 0$. A confined particle can never be at rest (consistent with the uncertainty principle). Nodes: ψ_n

has n − 1 nodes between the walls, and more nodes mean more curvature and more energy. Spacing: $E_{n+1} - E_n = (2n + 1)h^2/8mL^2$. Levels spread apart for higher n and for smaller L or m; for a macroscopic mass the spacing is immeasurably small, which is why we see continuous energy in everyday life.

Worked example: electron in a 1.0 nm box

```
E₁ = h²/8mL² = (6.626×10⁻³⁴)² / (8 × 9.109×10⁻³¹ × (10⁻⁹)²)
   = 6.03×10⁻²⁰ J = 0.376 eV
E₂ = 4E₁ = 1.50 eV      E₃ = 9E₁ = 3.38 eV
n = 1 → 2 absorption: ΔE = 3E₁ = 1.13 eV → λ = 1240/1.13 ≈ 1100 nm (near-infrared)
```

Probability: where is the particle?

$|\psi_n|^2$ is the probability density. For n = 1 the particle is most likely at the centre and never at the walls. Probability of being in the left quarter (0 to L/4):

```
P = (2/L)∫₀^(L/4) sin²(πx/L) dx = [x/L − sin(2πx/L)/2π] from 0 to L/4
  = 1/4 − 1/(2π) ≈ 0.091   (a classical particle would give 0.25)
```

For large n the density flattens to the classical value 1/L (the correspondence principle). Why it matters in chemistry: treating the π electrons of a conjugated chain as particles in a box (L ≈ chain length) predicts that longer chains have smaller gaps and absorb at longer wavelengths, the same reason quantum dots change colour with size.

```
Eₙ = n²h²/8mL²    ψₙ = √(2/L) sin(nπx/L)    ΔE = (2n+1)h²/8mL²
```

The hydrogen atom: full solution

S6–S8 · Lecture & Tutorial

The hydrogen atom is the one atom whose Schrödinger equation can be solved exactly. The potential comes from Coulomb attraction between the electron and the nucleus.

```
V(r) = −e²/(4πε₀r)
```

Because V depends only on the distance r, spherical polar coordinates (r, θ, φ) are natural, and the equation separates into independent pieces.

Step-by-step solution

1. **Separate the variables.** Write $\psi(r,\theta,\varphi) = R(r)\cdot\Theta(\theta)\cdot\Phi(\varphi) = R(r)\cdot Y(\theta,\varphi)$. Substituting into $H\psi = E\psi$ and dividing by RY gives one side depending only on r and the other only on angles. Each must equal a constant, called $l(l+1)$.

2. **Azimuthal equation.** The φ part is $d^2\Phi/d\varphi^2 = -m_l^2\Phi$ with solution $\Phi = e^{(im_l\varphi)}$. Going once round the axis ($\varphi \to \varphi + 2\pi$) must return the same value, so m_l must be an integer: 0, ±1, ±2, …

3. **Polar equation.** The θ equation has finite solutions only if l is a non-negative integer and $|m_l| \le l$. Together $\Theta\cdot\Phi = Y_l^{m_l}(\theta,\varphi)$ are the spherical harmonics, which fix the shape of the orbital.

4. **Radial equation.** The remaining equation contains the Coulomb term plus the centrifugal term $l(l+1)\hbar^2/2mr^2$. It has acceptable (finite, normalizable) solutions only ifThis is where the principal quantum number n comes from.

```
E = −13.6 eV / n²      n = 1, 2, 3, …      and l ≤ n − 1
```

The quantum numbers

n (size and energy): 1, 2, 3 … l (shape): 0 to n − 1, labelled s, p, d, f. m_l (orientation): −l to +l, giving 2l + 1 orbitals. Hence n = 2 has one 2s and three 2p orbitals; the total number of orbitals in shell n is n^2. (Spin m_s = ±½ is a separate fourth number.) All l values of a given n share the same energy in hydrogen (degenerate).

The 1s orbital in detail

```
ψ1s = (1/√(πa0³)) e^(−r/a0)      a0 = 4πε0ħ²/me² = 52.9 pm
```

It has no angular dependence (a sphere) and is largest at the nucleus. But the chance of finding the electron at distance r in a thin shell is the radial distribution function, $P(r) = 4\pi r^2\psi^2$, which includes the growing shell volume:

```
P(r) = (4/a0³) r² e^(−2r/a0)
dP/dr = 0  →  2r − 2r²/a0 = 0  →  r = a0
```

So the most probable distance is the Bohr radius, even though ψ itself peaks at r = 0.

Nodes and shapes

Radial nodes (spheres where ψ = 0) number n − l − 1, and angular nodes (planes or cones) number l. Total nodes = n − 1. 2s has one radial node at $r = 2a_0$. 2p has one angular node (a plane through the nucleus), giving two lobes of opposite sign. 3d has two angular nodes (four lobes, mostly).

Energy example

```
n = 2 → 1:  ΔE = 13.6(1 − 1/4) = 10.2 eV
λ = 1240 eV·nm / 10.2 eV = 121.6 nm   (Lyman-α line, ultraviolet)
```

Tutorial: the significance of ψ

ψ itself has no direct physical meaning and may be negative or complex. $|\psi|^2$ is the probability density. The sign (phase) of ψ matters when orbitals overlap: same signs add (bonding), opposite signs cancel (antibonding), which leads straight into molecular orbitals.

Molecular orbitals and Hückel theory

S11–S13 · Lecture & Tutorial

A molecular orbital (MO) is a one-electron wavefunction spread across the whole molecule. The standard approach builds it from atomic orbitals (LCAO).

LCAO for H_2^+ and H_2

1. **Trial function.** ψ = c_A φ_A + c_B φ_B, with φ_A and φ_B the 1s orbitals on atoms A and B.

2. **Variation principle.** The best coefficients are those that minimize E = ∫ψHψ dτ / ∫ψ²dτ. Minimizing with respect to c_A and c_B gives the secular equations, which have a non-zero solution only if the secular determinant is zero:where α = H_AA = H_BB (Coulomb integral, the energy of an electron on one atom), β = H_AB (resonance integral, negative) and S = ∫φ_Aφ_B dτ (overlap).

```
| H_AA − E     H_AB − ES |
| H_AB − ES   H_BB − E   | = 0
```

3. **Solve.** For identical atoms (α − E)² = (β − ES)², givingBecause β < 0, ψ_+ is lower than α (electron density builds up between the nuclei), and ψ_- is higher than α, with a node between the nuclei. The antibonding rise is larger than the bonding drop.

```
E₊ = (α + β)/(1 + S)      ψ₊ = (φ_A + φ_B)/√(2 + 2S)   bonding
E₋ = (α − β)/(1 − S)      ψ₋ = (φ_A − φ_B)/√(2 − 2S)   antibonding
```

Rules and diatomic molecules

Orbitals combine when they have similar energy, enough overlap and the same symmetry. N atomic orbitals make N MOs. Fill from the lowest with two electrons each (Pauli) and Hund's rule for degenerate levels.

```
Bond order = ½ (electrons in bonding − electrons in antibonding)
```

For O_2, F_2 the order is σ2s < σ*2s < σ2p < π2p < π*2p < σ*2p. For Li_2 to N_2, s–p mixing puts π2p below σ2p. Examples:

```
He₂   B.O. = ½(2 − 2) = 0   does not exist
N₂   B.O. = ½(10 − 4) = 3   very strong, diamagnetic
O₂   B.O. = ½(10 − 6) = 2   two unpaired electrons in π* → paramagnetic
O₂⁺ 2.5   O₂⁻ 1.5   (bond length rises as order falls)
```

Hückel theory for π systems

For planar conjugated molecules, treat only the π electrons, build each π MO from carbon 2p orbitals, and make three simple assumptions: all Coulomb integrals = α, resonance integral β between bonded neighbours only, and S = 0 and no other interactions. Dividing the secular determinant by β and writing x = (α − E)/β gives a determinant with x on the diagonal and 1 for each bonded pair.

Butadiene (4 carbons in a chain)

```
| x 1 0 0 |
| 1 x 1 0 |
| 0 1 x 1 |  = 0   →   x⁴ − 3x² + 1 = 0   →   x² = (3 ± √5)/2
| 0 0 1 x |
x = ±1.618, ±0.618   so   E = α ± 1.618β,  α ± 0.618β
```

With 4 π electrons filling the two lowest MOs: E(π) = 2(α + 1.618β) + 2(α + 0.618β) = 4α + 4.472β. Two isolated ethylene double bonds would give 4α + 4β, so the delocalization energy = 0.472|β|. The lowest MO is $\psi_1 = 0.372\varphi_1 + 0.602\varphi_2 + 0.602\varphi_3 + 0.372\varphi_4$, with no nodes, so the central C–C bond has partial double-bond character. The HOMO is ψ_2 and the LUMO is ψ_3, so the HOMO–LUMO gap is 1.236|β|.

Benzene (ring of 6 carbons)

```
Eₖ = α + 2β cos(2πk/6),   k = 0…5
α + 2β   α + β (×2)   α − β (×2)   α − 2β
E(π) = 2(α+2β) + 4(α+β) = 6α + 8β
```

Three isolated double bonds would be 6α + 6β, so the delocalization energy is 2|β| (the aromatic stabilization). All six C–C bonds are equal, the lowest MO is a ring-wide delocalized cloud, and the HOMO is a degenerate pair.

Tutorial note

Always follow this sequence: draw the skeleton → write the determinant → solve for x → convert to E → fill electrons → sum the energy → compare with localized double bonds.

In the lab

Practice

Unit 2: Periodic Properties & Cheminformatics

How atoms behave across the table, and how molecules become data a computer can search and learn from.

Effective nuclear charge and Slater's rules

S1 · Lecture

In a many-electron atom, an outer electron is attracted by the nucleus (+Z) but repelled by the other electrons, which partly cancel the attraction. This is shielding. The net pull actually felt is the effective nuclear charge:

```
Zeff = Z − S      (S = shielding or screening constant)
```

Slater gave simple rules to estimate S, so Zeff can be calculated for any electron.

Slater's rules

1. **Write the configuration in groups.** (1s)(2s,2p)(3s,3p)(3d)(4s,4p)(4d)(4f)(5s,5p)… Electrons to the right of the one considered do not shield it.

2. **For an s or p electron (ns, np):**

```
Same group:                  0.35 each (0.30 for 1s)
Shell n − 1:                  0.85 each
Shell n − 2 and lower:        1.00 each
```

3. **For a d or f electron (nd, nf):**

```
Same group:                  0.35 each
All groups to the left:       1.00 each
```

4. **Add up S and compute Zeff = Z − S.**

Worked examples

```
Na, 3s electron (Z = 11; 1s² 2s²2p⁶ 3s¹)
  S = 8(0.85) + 2(1.00) = 8.8          Zeff = 2.2

Cl, 3p electron (Z = 17; 1s² 2s²2p⁶ 3s²3p⁵)
  S = 6(0.35) + 8(0.85) + 2(1.00) = 10.9   Zeff = 6.1

Zn (Z = 30), 4s electron:  S = 0.35 + 18(0.85) + 10(1.00) = 25.65   Zeff = 4.35
Zn (Z = 30), 3d electron:  S = 9(0.35) + 18(1.00) = 21.15           Zeff = 8.85
```

What they show. Across a period (Na 2.2 → Cl 6.1), Zeff rises because each new proton adds +1 but each new electron in the same shell shields only 0.35, so atoms shrink and hold electrons more tightly. Down a group, Zeff stays roughly constant but the outer electron sits in a larger shell, so size grows. In Zn the 3d electron feels more charge than the 4s, which is why 4s electrons are lost first when transition metals ionize.

Slater's effective principal quantum number n* (n = 1, 2, 3, 4, 5, 6 → n* = 1, 2, 3, 3.7, 4.0, 4.2) allows a rough energy: $E = -13.6\,(Zeff/n^*)^2$ eV.

Ionization energy, electron affinity, electronegativity, polarizability

S2 · Lecture

Ionization energy (IE)

Energy needed to remove the outermost electron from a gaseous atom: $M(g) \rightarrow M^+(g) + e^-$. It rises with Zeff and falls with distance. First IE (eV) for period 2:

```
Li 5.39  Be 9.32  B 8.30  C 11.26  N 14.53  O 13.62  F 17.42  Ne 21.56
```

General rise: Zeff increases. B below Be: B's electron is in a 2p orbital, higher in energy and shielded by 2s. O below N: O has a paired 2p electron, and pair repulsion makes it easier to remove. Down a group (Li 5.39, Na 5.14, K 4.34 eV): larger radius, weaker pull. Successive IEs always increase, with a huge jump after the valence shell is emptied. Mg: 738, 1451, then 7733 kJ/mol, which shows that Mg is +2.

Electron affinity (EA)

Energy change when a gaseous atom gains an electron: $X(g) + e^- \rightarrow X^-(g)$. Large negative values (energy released) are found for halogens. Cl (−349 kJ/mol) is more negative than F (−328) because F is so small that the extra electron suffers strong repulsion. N is near zero (the new electron must pair in a half-filled p subshell) and noble gases are positive (new electron needs a new shell).

Electronegativity (χ)

The ability of an atom in a bond to attract the shared electrons. Pauling: derived from bond energies (F 3.98, O 3.44, Cl 3.16, N 3.04, C 2.55, H 2.20, Na 0.93). Mulliken: the average of IE and EA. Allred–Rochow: based on Zeff and covalent radius r.

$$\chi(\text{Mulliken}) = (IE + EA)/2 \qquad \chi(\text{A–R}) = 0.359\, Zeff/r^2 + 0.744$$

Use: a difference above about 1.7 suggests ionic bonding; 0.4 to 1.7 polar covalent; below 0.4 nearly pure covalent (C–H).

Polarizability (α)

How easily the electron cloud is distorted by an electric field (induced dipole $\mu = \alpha E$). It grows with atomic volume ($\approx r^3$) and with the number of loosely held electrons, so it increases down a group and decreases across a period. Consequences: stronger London dispersion forces, hence higher boiling points down the noble-gas group (He 4 K, Ar 87 K, Xe 165 K), and by Fajans' rules a large, easily polarized anion (I^-) combined with a small, highly charged cation gives a more covalent bond.

Tutorial: Zeff problems with solutions

S3 · Tutorial

1. **C 2p electron (Z = 6; $1s^2\ 2s^2 2p^2$).** S = 3(0.35) + 2(0.85) = 2.75, so Zeff = 3.25.
2. **F 2p electron (Z = 9).** S = 6(0.35) + 2(0.85) = 3.80, so Zeff = 5.20. Compare with C: this is the reason F is much smaller and has a larger IE.
3. **Mg 3s electron (Z = 12).** S = 0.35 + 8(0.85) + 2(1.00) = 9.15, so Zeff = 2.85.

4. **K 4s electron (Z = 19).** S = 8(0.85) + 10(1.00) = 16.8, so Zeff = 2.2; with n* = 3.7 this explains why K's IE is lower than Na's.

5. **Method.** Write the configuration → group the orbitals → pick the right rule (s/p vs d/f) → count → Zeff = Z − S → compare with neighbours.

Check yourself: predict which of O^{2-}, F^{-}, Na^{+} has the largest radius. All have 10 electrons, so the one with the smallest Z has the smallest Zeff, and O^{2-} is largest.

Practical: hardness of water by EDTA titration

S4–S5 · Practice

Idea. Hardness comes from dissolved Ca^{2+} and Mg^{2+} salts. EDTA (H_4Y) is a six-point (hexadentate) chelating agent that wraps around a metal ion and binds it 1:1 whatever the charge: $M^{2+} + H_2Y^{2-} \rightarrow MY^{2-} + 2H^{+}$. Because H^{+} is released, the titration is carried out in a pH 10 buffer (NH_3/NH_4Cl).

1. **Indicator.** Eriochrome Black T (EBT) is blue when free at pH 10, and wine-red when bound to Ca^{2+}/Mg^{2+}. EDTA binds the metal more strongly than EBT does, so at the end point it takes the metal from EBT and the colour changes from wine red to blue.

2. **Standardize EDTA.** Titrate a known solution (e.g. standard $CaCO_3$ dissolved in HCl, or $ZnSO_4$) to find its exact molarity.

3. **Titrate the water.** Pipette 50 mL of water, add 2 mL buffer and a few drops of EBT, and titrate with EDTA until the colour just turns blue. Repeat until concordant (±0.1 mL).

4. **Calculate.**

```
Hardness (ppm as CaCO₃) = M(EDTA) × V(EDTA, mL) × 100.09 × 1000 / V(sample, mL)
Example: 0.01 M × 10.0 mL × 100.09 × 1000 / 50 mL = 200 ppm
```

Interpretation (approximate, WHO): under 60 ppm soft; 60–120 moderate; 120–180 hard; over 180 very hard. Temporary hardness (bicarbonates) is removed by boiling; permanent hardness (chlorides, sulfates) needs softening such as ion exchange. Sources of error: wrong pH, missing the faint end point, and traces of Fe or Cu that block the indicator.

Basics of cheminformatics

S6 · Lecture

Cheminformatics applies computer science to chemical problems: storing structures, searching them, calculating properties, and predicting behaviour of molecules without having to make each in the lab. A typical question: "out of 10 million candidate molecules, which few are likely to bind this protein and not be toxic?"

```
Molecule → computer-readable representation → descriptors → model → prediction → experiment
```

The key idea is that a molecule is a graph: atoms are nodes, bonds are edges, and labels carry atom type and bond order. Most algorithms work on this graph or numbers derived from it.

Descriptors

Numbers that describe a molecule: molecular weight, logP (oil/water preference), hydrogen-bond donors (HBD) and acceptors (HBA), polar surface area (TPSA), rotatable bonds, ring counts. Lipinski's rule of five (oral drug-likeness): MW ≤ 500, logP ≤ 5, HBD ≤ 5, HBA ≤ 10. Aspirin (MW 180, HBD 1, HBA 4) passes easily.

Where it is used

Virtual screening and docking, QSAR (quantitative structure–activity relationships), ADMET prediction (absorption, distribution, metabolism, excretion, toxicity), reaction and retrosynthesis planning, and materials discovery. Similar structure often means similar property (the "similarity principle"), which is the foundation of the whole field.

Representing molecules in two and three dimensions

S7, S8 · Lecture & Tutorial

Ways to represent one molecule

Molecular formula (C_2H_6O) is ambiguous: ethanol and dimethyl ether both match. A structural graph fixes connectivity. A connection table lists atoms and bonds; the adjacency matrix for the heavy atoms of ethanol (C_1–C_2–O_3) is

```
     C1 C2 O3
C1 [ 0  1  0 ]
C2 [ 1  0  1 ]
O3 [ 0  1  0 ]
```

Text strings (SMILES, InChI), bit-vector fingerprints (does the molecule contain substructure X?) and 3D coordinates complete the list. Fingerprint similarity uses the Tanimoto coefficient T = c/(a + b − c) where a and b are the bits set in each molecule and c those set in both; 1 is identical, 0 is no overlap.

Why 3D matters

Binding, reactivity and spectra depend on shape: enantiomers have identical 2D graphs but different 3D shapes (and different biological effects). Flexible molecules exist as conformers by rotating about single bonds (torsion angles), and software searches for the lowest-energy ones using a force field such as MMFF94.

Coordinates and geometry

A 3D structure is a list of atoms with x, y, z in ångströms (XYZ format: atom count, title, then one line per atom). Bond lengths and angles are computed from coordinates:

```
d = √((x₂−x₁)² + (y₂−y₁)² + (z₂−z₁)²)
cos θ = (u·v)/(|u||v|)   where u and v are the bond vectors from the central atom
```

Example, water: O (0, 0, 0), H_1 (0.757, 0.586, 0), H_2 (−0.757, 0.586, 0).

```
d(O–H) = √(0.757² + 0.586²) = 0.957 Å
u·v = −0.573 + 0.343 = −0.230,  |u||v| = 0.916
cos θ = −0.251  →  θ = 104.5°
```

Other 3D formats: MOL/SDF (coordinates plus bonds), MOL2, PDB (proteins). Internal coordinates (Z-matrix) describe geometry through bond lengths, angles and torsions instead. Practice (S9–S10): sketch a molecule, export a 2D MOL file, generate a 3D conformer, optimize it, and measure one bond length and one angle.

The SMILES format

S11 · Lecture

SMILES (Simplified Molecular Input Line Entry System) writes a molecular graph as a short text string by walking through the atoms.

1. **Atoms.** Organic-subset atoms (B, C, N, O, P, S, F, Cl, Br, I) are written plainly, with hydrogen counts filled in from normal valence. Others go in brackets: [Na+], [NH4+], [O-], [13C].
2. **Bonds.** Single is implied (or −), double =, triple #, aromatic bonds are implied between lowercase atoms.
3. **Branches.** Parentheses: CC(C)C is isobutane.
4. **Rings.** Match digits: the two atoms labelled 1 are bonded. C1CCCCC1 is cyclohexane; benzene is c1ccccc1 (lowercase = aromatic).
5. **Stereo.** @ and @@ give chirality (viewed from the first neighbour, the others go anticlockwise or clockwise); / and \ give double-bond E/Z.
6. **Disconnected parts.** A dot, e.g. [Na+].[Cl-].

```
methane          C                    ethanol        CCO
acetic acid      CC(=O)O              acetonitrile   CC#N
acetone          CC(=O)C              L-alanine      N[C@@H](C)C(=O)O
paracetamol      CC(=O)Nc1ccc(O)cc1   aspirin        CC(=O)Oc1ccccc1C(=O)O
caffeine         Cn1cnc2c1c(=O)n(C)c(=O)n2C
```

Reading acetic acid CC(=O)O: C (methyl) → C (carbonyl carbon) with a branch (=O) double-bonded oxygen → O (hydroxyl). Non-uniqueness: CCO and OCC are both ethanol. Software applies a canonicalization algorithm so each molecule has one canonical SMILES, which is crucial for finding duplicates in databases. InChI is a related standard identifier that is unique by design.

Standard software tools

S12 · Lecture

RDKit (open-source Python/C++): reads SMILES and files, draws structures, calculates descriptors and fingerprints, performs substructure search and generates 3D conformers. Open Babel: converts between over 100 chemical file formats. Avogadro, PyMOL, ChimeraX: build and visualize in 3D. MarvinSketch, ChemDraw: drawing. AutoDock Vina: docking of ligands into proteins. ORCA, Gaussian: quantum-chemical calculations. scikit-learn, PyTorch: machine-learning models.

```
from rdkit import Chem
from rdkit.Chem import Descriptors, AllChem
m = Chem.MolFromSmiles('CC(=O)Oc1ccccc1C(=O)O')   # aspirin
print(Descriptors.MolWt(m))                        # 180.16
mh = Chem.AddHs(m)
AllChem.EmbedMolecule(mh)                          # make 3D coordinates
AllChem.MMFFOptimizeMolecule(mh)                   # relax with force field

# Open Babel, command line:  obabel -:"CCO" -O ethanol.sdf --gen3d
```

Chemical databases and how to use them

S13 · Tutorial

PubChem: tens of millions of compounds with structures, properties and bioassays (aspirin is CID 2244). ChEMBL: curated bioactivity of drug-like molecules against targets. ZINC: purchasable compounds, ready-made for docking. DrugBank: approved drugs and targets. PDB: 3D structures of proteins and complexes. CSD: small-molecule crystal structures. NIST WebBook: thermochemical and spectral data.

1. **Search by:** name, formula, SMILES or InChI, substructure (contains this ring), or similarity (Tanimoto above a threshold).
2. **Download** as SDF or CSV for analysis.
3. **Clean the data.** Standardize SMILES, strip salts and solvents, merge duplicates, make units consistent, and check outliers. Poor data gives poor models, regardless of the algorithm.
4. **Cite the source and its version.** Databases are updated over time.

Practical: AI classification model for drug toxicity

S14–S15 · Practice

Goal: given a molecule, predict whether it is toxic. This is binary classification, trained on molecules with known labels (for example the public Tox21 data, about 8,000 compounds over 12 assays).

1. **Collect and clean.** Load SMILES and labels, standardize, remove duplicates and invalid structures.
2. **Featurize.** Morgan (circular) fingerprints: each atom's neighbourhood up to a radius (usually 2) is hashed to a position in a 2048-bit vector. Each molecule becomes a row of 0s and 1s. Alternative: RDKit descriptors or a graph neural network on the graph itself.
3. **Split.** Train/test (e.g. 80/20). A scaffold split puts different ring-system families in train and test, which gives an honest estimate for new chemistry.
4. **Train.** Logistic regression, random forest or gradient boosting.
5. **Evaluate.** Toxic molecules are rare, so accuracy alone misleads: a model that says "non-toxic" always can score 90%. Use recall, precision and AUC.

```
precision = TP/(TP+FP)       recall = TP/(TP+FN)
ROC-AUC: 0.5 = random guessing, 1.0 = perfect
```

6. **Interpret and limit.** Inspect which bits matter (they map back to substructures, such as nitro groups and reactive electrophiles). A model is reliable only for molecules similar to its training data, the applicability domain. It cannot replace experiments.

```
X = [AllChem.GetMorganFingerprintAsBitVect(m, 2, 2048) for m in mols]
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2, stratify=y)
clf = RandomForestClassifier(n_estimators=300).fit(X_tr, y_tr)
print(roc_auc_score(y_te, clf.predict_proba(X_te)[:, 1]))
```

Unit 3: Thermodynamics & Electrochemistry

Energy, entropy and free energy, then how they decide the voltage of a cell.

Basic terms and the four laws

S3 · Tutorial

Thermodynamics tracks energy changes in bulk matter. We first fix the vocabulary.

Basic terms

System: the part of the universe we study. Surroundings: everything else. Boundary: what separates them. Open system: exchanges matter and energy (boiling pot). Closed: exchanges energy only (sealed flask). Isolated: exchanges neither (ideal thermos). Homogeneous: one phase with uniform properties (salt solution). Heterogeneous: more than one phase (ice in water).

Extensive properties depend on amount (mass, volume, U, H, S, G); intensive ones do not (T, P, density). A state function depends only on the present state, not the path, so its change is final − initial. Heat q and work w are path functions. Processes: isothermal (T constant), isobaric (P), isochoric (V), adiabatic (q = 0), and reversible (infinitesimally slow, always at equilibrium).

The laws

1. **Zeroth law.** If A and B are each in thermal equilibrium with C, they are in equilibrium with each other. This defines temperature and makes thermometers possible.

2. **First law.** Energy is conserved. Using the convention that q and w are positive when they enter the system:

   ```
   ΔU = q + w
   ```

3. **Second law.** The entropy of an isolated system (or of system + surroundings) never decreases: ΔS(universe) ≥ 0. Heat does not flow spontaneously from cold to hot.

4. **Third law.** The entropy of a perfect crystal is zero at 0 K, so absolute entropies can be tabulated.

Thermodynamic functions: internal energy and enthalpy

S1 · Lecture

Internal energy U is the total kinetic and potential energy of the molecules in the system. Only changes ΔU can be measured. It changes through heat and work.

Work of expansion

A gas pushing against an external pressure does work w = −P_ext ΔV (negative: the system loses energy). For a reversible isothermal expansion of an ideal gas, integrating dw = −P dV with P = nRT/V:

```
w = −nRT ln(V₂/V₁)
```

Example: 1 mol expands isothermally at 298 K to double volume: w = −(1)(8.314)(298)(0.693) = −1.72 kJ. Since U of an ideal gas depends only on T, ΔU = 0 and q = +1.72 kJ.

Enthalpy

1. **At constant volume** no expansion work, so ΔU = q_v. Heat measured in a bomb calorimeter is ΔU.

2. **At constant pressure** ΔU = q_p − PΔV. Define enthalpy H = U + PV. Then ΔH = ΔU + PΔV = q_p. Heat measured in an open beaker is ΔH.

3. **Relating them for gases:** Example: $N_2 + 3H_2 \rightarrow 2NH_3$ has Δn = −2, so ΔH is lower than ΔU by 2RT ≈ 4.96 kJ at 298 K.

```
ΔH = ΔU + Δn(g)·RT
```

Heat capacity

C_V = (∂U/∂T)_V and C_P = (∂H/∂T)_P. For one mole of ideal gas H = U + RT, so differentiating with respect to T gives

```
C_P − C_V = R
```

Hess's law and enthalpies of formation

Because H is a state function, the enthalpy of a reaction is the same whether done in one step or several.

```
ΔH°rxn = ΣΔH°f(products) − ΣΔH°f(reactants)
CH₄ + 2O₂ → CO₂ + 2H₂O(l):
= −393.5 + 2(−285.8) − (−74.8) = −890.3 kJ/mol
```

Entropy: meaning and estimation

S2, S6, S8 · Lecture & Tutorial

Energy conservation cannot explain why processes have a direction. Entropy does.

Two definitions

Thermodynamic (Clausius): heat absorbed reversibly divided by temperature. Statistical (Boltzmann): S = k ln W, where W is the number of microscopic arrangements (microstates) matching the macroscopic state. More ways to arrange energy and particles means higher entropy, which is why gases > liquids > solids and mixtures > pure substances.

```
dS = q_rev / T        S = k ln W        k = 1.381×10⁻²³ J/K
```

How to estimate ΔS

1. **Phase change (T constant).** Ice melting: 6010 / 273.15 = 22.0 J K^{-1} mol^{-1}. Water boiling: 40660 / 373.15 = 109 J K^{-1} mol^{-1}. Vaporization is larger because the gas has far more microstates.

```
ΔS = ΔH(transition)/T
```

2. **Heating at constant pressure.** Using dq_rev = nC_P dT and integrating:Example: 1 mol of water from 298 to 348 K, C_P = 75.3 J/K: ΔS = 75.3 × ln(348/298) = 11.7 J/K.

```
ΔS = nC_P ln(T₂/T₁)
```

3. **Ideal gas expansion (isothermal).** Doubling the volume of 1 mol: 8.314 × 0.693 = 5.76 J/K.

```
ΔS = nR ln(V₂/V₁)
```

4. **Chemical reaction.** Absolute S° values exist because of the third law. Sign guide: more gas moles after reaction means ΔS > 0.

```
ΔS°rxn = ΣS°(products) − ΣS°(reactants)
```

Entropy of the universe

A spontaneous process has ΔS_univ = ΔS_sys + ΔS_surr > 0. Heat given to the surroundings at constant pressure is −ΔH_sys, so ΔS_surr = −ΔH_sys/T, which sets up free energy.

Tutorial: enthalpy estimation from bond energies

Bonds broken (energy in) minus bonds formed (energy out). $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O(g)$:

```
Broken: 4(C–H 413) + 2(O=O 498) = 2648 kJ
Formed: 2(C=O 799) + 4(O–H 463) = 3450 kJ
ΔH ≈ 2648 − 3450 = −802 kJ/mol   (tabulated: −802.3)
```

Gibbs free energy: derivation and use

S2, S6 · Lecture

1. **Start from the second law.** ΔS_univ = ΔS_sys + ΔS_surr > 0 for a spontaneous process.

2. **Insert ΔS_surr = −ΔH_sys/T (constant T and P):**

```
ΔS_univ = ΔS_sys − ΔH_sys/T
```

3. **Multiply by −T and name the result.** So ΔG = −TΔS_univ, and a spontaneous change at constant T and P has ΔG < 0. ΔG = 0 means equilibrium.

```
−TΔS_univ = ΔH − TΔS_sys ≡ ΔG
G = H − TS
```

Signs: ΔH < 0 and ΔS > 0 means spontaneous at all T. ΔH > 0 and ΔS < 0 means never. ΔH < 0, ΔS < 0 is spontaneous at low T. ΔH > 0, ΔS > 0 is spontaneous at high T (crossover T = ΔH/ΔS).

Free energy and equilibrium

```
ΔG = ΔG° + RT ln Q
At equilibrium ΔG = 0 and Q = K:   ΔG° = −RT ln K
```

Large negative ΔG° means K ≫ 1. Also, ΔG equals the maximum useful (non-expansion) work obtainable: w_max = ΔG, the link to batteries below. Estimation: ΔG°rxn = ΣΔG°f(products) − ΣΔG°f(reactants), or ΔH° − TΔS°.

Case study: ammonia synthesis (Haber–Bosch)

S9–S10 · Practice

```
N₂ + 3H₂ ⇌ 2NH₃
```

1. **ΔH°.** −92.2 kJ (exothermic).

2. **ΔS°.** 2(192.5) − 191.6 − 3(130.7) = −198.7 J/K (4 gas moles become 2).

3. **ΔG° at 298 K.** Very favourable at room temperature.

```
−92.2 − 298(−0.1987) = −33.0 kJ
K = e^(33000/(8.314×298)) ≈ 6×10⁵
```

4. **Crossover temperature.** Above this ΔG° > 0. At 700 K: ΔG° = −92.2 + 139.1 = +46.9 kJ and K ≈ 3×10^{-4}.

```
T = ΔH/ΔS = 92200/198.7 ≈ 464 K
```

Why the plant uses 400–500 °C anyway: at room temperature the rate is essentially zero. High T gives speed but lowers the equilibrium yield (Le Chatelier). The compromise: ~450 °C, 150–300 atm (high P favours fewer gas moles) and an iron catalyst. A catalyst speeds the approach to equilibrium but does not change ΔG° or K. Unreacted gas is recycled, and NH_3 is condensed out. Practice task: compute ΔG° and K at several temperatures and plot them.

Free energy and EMF of cells

S7, S12 · Lecture

A galvanic cell separates a spontaneous redox reaction into two half-cells so that electrons flow through an external wire. Anode: oxidation (negative terminal). Cathode: reduction (positive). A salt bridge completes the circuit. Daniell cell notation: Zn | Zn^{2+} (1 M) ‖ Cu^{2+} (1 M) | Cu.

```
Anode:   Zn → Zn²⁺ + 2e⁻     E° = −0.76 V (as reduction)
Cathode: Cu²⁺ + 2e⁻ → Cu     E° = +0.34 V
E°cell = E°(cathode) − E°(anode) = 0.34 − (−0.76) = 1.10 V
```

Link between ΔG and E

1. **Electrical work.** Moving n moles of electrons through potential E does work nFE (F = 96485 C/mol, Faraday constant).

2. **Maximum work equals ΔG.** Daniell: ΔG° = −2(96485)(1.10) = −212 kJ. Spontaneous cell means E > 0.

```
ΔG = −nFE        ΔG° = −nFE°
```

3. **Equilibrium constant.** Combine ΔG° = −RT ln K with ΔG° = −nFE°:Daniell: log K = 2(1.10)/0.0592 = 37.2, so K ≈ 10^{37}.

```
E° = (RT/nF) ln K = (0.0592/n) log K   (298 K)
```

Standard potentials are measured against the standard hydrogen electrode (Pt | H_2 1 bar | H^+ 1 M), defined as 0 V. A more positive E° means a stronger oxidizing agent.

The Nernst equation and electrode potentials

S11, S13 · Lecture & Tutorial

1. **Start with the free energy of a reaction.**

$$\Delta G = \Delta G^\circ + RT \ln Q$$

2. **Replace ΔG with −nFE.**

$$-nFE = -nFE^\circ + RT \ln Q$$

3. **Divide by −nF.**

$$E = E^\circ - (RT/nF) \ln Q$$
$$E = E^\circ - (0.0592/n) \log Q \quad \text{at } 25\ ^\circ C$$

Example: Daniell cell with $[Zn^{2+}] = 0.01$ M and $[Cu^{2+}] = 1$ M: Q = 0.01, so E = 1.10 − (0.0592/2) log(0.01) = 1.10 + 0.059 = 1.159 V. As the reaction proceeds, Q → K and E → 0 (a dead battery).

Tutorial: electrode potential expressions

For a half-reaction Ox + ne^- → Red the potential is E = E° − (0.0592/n) log([Red]/[Ox]). Pure solids and liquids count as 1.

Metal–metal ion, $M^{n+} + ne^- \rightarrow M$:	$E = E^\circ + (0.0592/n) \log[M^{n+}]$
Hydrogen, $2H^+ + 2e^- \rightarrow H_2$ (1 bar):	$E = -0.0592$ pH
Ag/AgCl, $AgCl + e^- \rightarrow Ag + Cl^-$:	$E = 0.222 - 0.0592 \log[Cl^-]$
Calomel (sat. KCl):	$E \approx 0.241$ V (reference electrode)
Redox, Pt | Fe^{3+}, Fe^{2+}:	$E = 0.77 + 0.0592 \log([Fe^{3+}]/[Fe^{2+}])$
Glass electrode:	$E = \text{const} - 0.0592$ pH

Concentration cells

Same electrode, different concentrations: E° = 0, so E = (0.0592/n) log(c_2/c_1). The cell runs until the two concentrations equalize, a pure entropy-driven process.

Applications

pH measurement; finding K and K_sp from E°; potentiometric titrations; batteries and fuel cells; corrosion (rusting is a galvanic process in which iron is the anode); electroplating and electrolysis (the reverse process, needing E > E°cell).

Potentiometric titration of Fe^{2+} with $K_2Cr_2O_7$

S14–S15 · Practice

$$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$

1. **Set up.** Pt indicator electrode and a calomel reference in the acidified Fe^{2+} sample. Add dichromate in small portions and record the potential after each.

2. **What happens.** Before the end point E = 0.77 + 0.0592 log($[Fe^{3+}]/[Fe^{2+}]$), which rises slowly. At the equivalence point nearly all Fe^{2+} is gone and E jumps sharply towards the dichromate couple (E° = 1.33 V).

3. **Find the end point.** Plot E vs V, then ΔE/ΔV vs V (peak at the end point), or $\Delta^2E/\Delta V^2$ (crosses zero).

4. **Calculate.** The Fe^{2+} concentration follows from the equivalence volume. Advantage over an indicator: works with coloured or turbid solutions.

```
N₁V₁ = N₂V₂  (equivalent weight of K₂Cr₂O₇ = M/6)
```

Related tasks (S4–S5): viva on basics. (S9–S10): Haber–Bosch Gibbs-energy calculation above.

Unit 4: Materials Chemistry

Magnets, superconductors, semiconductors and nanomaterials, and why small things behave differently.

Magnetic materials

S1 · Lecture

Magnetism comes from electrons: each has a spin magnetic moment (and orbital motion adds a little). Atoms with all electrons paired have no net moment; atoms with unpaired electrons are tiny magnets. How these moments behave together gives five types. The magnetization M of a material in a field H is measured by the susceptibility $\chi = M/H$.

1. **Diamagnetic.** All electrons paired (Cu^{+}, Zn^{2+}, water, Bi). The field induces a weak opposing moment, so χ is small and negative, and the material is slightly repelled.

2. **Paramagnetic.** Unpaired electrons, but the moments point randomly due to thermal motion. A field aligns a few, so χ is small and positive (Curie law $\chi = C/T$). The spin-only moment is $\mu = \sqrt{n(n+2)}\ \mu B$, where n is the number of unpaired electrons: Cu^{2+} (n = 1) gives 1.73 μB; high-spin Fe^{3+} (n = 5) gives $\sqrt{35} = 5.92\ \mu B$.

3. **Ferromagnetic.** Strong coupling aligns neighbouring moments in the same direction within regions called domains (Fe, Co, Ni). χ is very large, and the material keeps magnetization after the field is removed. Above the Curie temperature T_C (Fe 1043 K, Co 1388 K, Ni 627 K) heat destroys the order and the material becomes paramagnetic: $\chi = C/(T - T_C)$.

4. **Antiferromagnetic.** Neighbouring moments point in opposite directions and cancel (MnO, Cr). Order disappears above the Néel temperature.

5. **Ferrimagnetic.** Opposing moments of unequal size leave a net moment (magnetite Fe_3O_4, ferrites MFe_2O_4). They are magnetic but insulating, so they are useful at high frequency.

The hysteresis loop

Magnetizing a ferromagnet, reversing the field and repeating traces a loop of M against H. Remanence is the magnetization left at H = 0; coercivity is the reverse field needed to bring M to zero; the loop area is energy lost as heat per cycle.

```
Soft magnets: narrow loop, low coercivity, low loss  → transformer cores, inductors,
electromagnets (silicon steel, soft ferrites)
Hard magnets: wide loop, high coercivity, strong remanence → permanent magnets,
motors, speakers (Nd₂Fe₁₄B, SmCo₅, alnico), hard-disk media
```

Superconducting materials

S2 · Lecture

In 1911 Kamerlingh Onnes found that mercury's resistance dropped abruptly to zero at 4.2 K. Below a critical temperature T_c a superconductor has two defining properties.

1. **Zero resistance.** A current in a superconducting ring flows for years without any measurable decay.

2. **Meissner effect.** The material expels magnetic field from its interior (perfect diamagnetism, $\chi = -1$). This is different from a merely perfect conductor and is what makes magnets levitate above a superconductor.

Superconductivity is destroyed if $T > T_c$, if the magnetic field exceeds a critical field H_c, or if the current exceeds a critical current. Approximately $H_c(T) = H_c(0)[1 - (T/T_c)^2]$.

Type I and Type II

Type I (pure metals such as Hg, Pb): the superconducting state collapses abruptly at H_c. Type II (alloys and ceramics: Nb_3Sn, YBCO): between H_{c1} and H_{c2} the field penetrates in thin tubes (vortices) while the rest remains superconducting, so they tolerate very high fields and are the ones used in practice.

BCS theory (low-temperature superconductors)

1. **An electron passing through the lattice pulls the positive ions slightly toward it.** This leaves a region of extra positive charge.

2. **A second electron is attracted to that region.** Two electrons with opposite momentum and spin form a bound Cooper pair (via lattice vibrations, phonons).

3. **Pairs behave as bosons.** They condense into one coherent quantum state that flows without scattering. An energy gap $\Delta \approx 1.76\, k_B T_c$ separates the paired state from ordinary electrons, so small vibrations cannot break it.

Supporting evidence is the isotope effect, $T_c \propto M^{(-1/2)}$: heavier isotopes of the lattice have lower T_c. High-T_c cuprates are not fully explained by BCS.

```
Hg 4.2 K   Nb 9.2 K   NbTi 9.5 K   Nb₃Sn 18 K   MgB₂ 39 K   YBa₂Cu₃O₇ (YBCO) 92 K
Hg-cuprates ~133 K
```

YBCO works above the boiling point of liquid nitrogen (77 K), which is cheap. Applications: MRI magnets (NbTi), the LHC accelerator magnets, maglev trains, SQUID magnetometers, lossless power cables and fault-current limiters. The main limitation is the cost of cooling and the brittleness of ceramics.

Band structure of solids and the role of doping

S3 · Tutorial

How bands form

Take Unit 1's idea: two atomic orbitals combine into a bonding and an antibonding MO. With N atoms, N orbitals combine into N molecular levels spread over an energy width of a few eV. In a solid $N \approx 10^{23}$, so the levels are so closely spaced that they form a continuous band. Each atomic level (3s, 3p …) becomes its own band, and gaps between bands are energies with no allowed states. Bands made from the occupied orbitals are the valence band (VB); those from empty ones are the conduction band (CB); the gap is E_g.

Metals, insulators, semiconductors

```
Metal:          band partly filled (Na, 3s half full) or VB and CB overlap (Mg).
Electrons can move at once → high conductivity.
Insulator:      VB full, CB empty, E_g large (> ~4 eV; diamond 5.5 eV). No carriers.
```

```
Semiconductor:  same picture but small E_g (Ge 0.67, Si 1.12, GaAs 1.42 eV). Heat or light can excite electrons.
```

The Fermi level is the highest occupied energy at 0 K. When an electron jumps to the CB it leaves a hole in the VB, and both carry current. Temperature raises conductivity of a semiconductor but lowers it in a metal (more lattice scattering).

Doping

Adding a trace of a different element (one atom in 10^6 to 10^9) greatly changes conductivity.

1. **n-type.** Phosphorus (5 valence electrons) replaces a Si atom (4). Four electrons bond; the fifth is loosely held and sits in a donor level about 0.045 eV below the CB, so it is excited into the CB even at room temperature. Majority carriers: electrons.

2. **p-type.** Boron (3 valence electrons) leaves a missing bond, a hole. An acceptor level just above the VB captures VB electrons, leaving mobile holes. Majority carriers: holes.

3. **Quantitative.** Mass-action law: $n \cdot p = n_i^2$. Pure Si at 300 K has $n_i \approx 10^{10}$ cm^{-3}. For 10^{16} cm^{-3} of P: $n \approx 10^{16}$ and $p = (10^{10})^2/10^{16} = 10^4$ cm^{-3}. Conductivity:which is about 10^5 times that of pure Si.

```
σ = q(nμₑ + pμₕ) = (1.6×10⁻¹⁹)(10¹⁶)(1350) ≈ 2.2 S/cm   (electron mobility of Si ≈ 1350 cm²/V s)
```

Semiconductor materials

S6 · Lecture

Intrinsic semiconductors are pure; their conductivity depends only on thermal excitation across the gap, with the number of carriers given by

```
n ∝ exp(−E_g / 2kT)      σ = σ₀ exp(−E_g / 2kT)
```

Extrinsic semiconductors are doped. Common materials: Si and Ge (elemental); GaAs, InP, GaN (III–V compounds); CdS, ZnO (II–VI).

Direct and indirect gaps

In a direct semiconductor (GaAs, GaN) the CB minimum is directly above the VB maximum, so an electron falls back and emits a photon efficiently, which is why they are used in LEDs and lasers. Si is indirect, so recombination mainly produces heat, which makes Si poor for light emission but still the standard for chips and solar cells. The emitted wavelength follows $\lambda(\text{nm}) = 1240/E_g(\text{eV})$; GaN (3.4 eV) emits in the near-UV/blue, which led to white LED lighting.

The p–n junction

1. **Join p- and n-type.** Electrons diffuse from n to p and holes from p to n, leaving fixed charged ions behind.

2. **A depletion region forms.** It has an internal electric field that stops further diffusion.

3. **Forward bias (p to +).** The barrier shrinks and current flows. Reverse bias widens the barrier and almost no current flows, so the junction is a one-way valve (a diode).

4. **Devices.** LED (recombination emits light), solar cell (light creates pairs which the field separates), transistor (two junctions, a switch and amplifier).

Making semiconductors: silicon is purified by zone refining to better than one impurity in 10^9, grown as single crystals (Czochralski method), and then doped by diffusion or ion implantation.

Introduction to nanomaterials

S7 · Lecture

Nanomaterials have at least one dimension between 1 and 100 nm. For scale: a human hair is about 80,000 nm wide, a DNA helix 2 nm, a hydrogen atom 0.1 nm. A nanometre is one billionth of a metre.

Why small is different

Two effects switch on at this size. (1) Surface effects: a large fraction of the atoms sit on the surface, and these are less strongly bonded and more reactive. (2) Quantum effects: electrons are confined, so energy levels and the band gap depend on size (see quantum dots).

How they are made

```
Top-down:  start from bulk and break it down: ball milling, lithography, etching,
laser ablation
Bottom-up: build from atoms and molecules: sol-gel, co-precipitation, chemical vapour
deposition, hydrothermal, self-assembly
```

Bottom-up chemistry gives better control of size and uniformity. In solution, a capping agent (citrate, thiols, polymers) sticks to the particle surface to stop growth and prevent clumping.

Properties of nanomaterials

S8 · Tutorial

1. **Surface-to-volume ratio.** For a sphere $SA/V = 4\pi r^2/(4/3\ \pi r^3) = 3/r$, so it rises as r falls. Roughly, with atoms 0.3 nm across, the fraction of atoms at the surface is $1 - ((r - 0.3)/r)^3$:Consequence: much higher catalytic activity per gram.

```
10 nm particle (r = 5 nm):   ≈ 17 % of atoms on the surface
3 nm particle  (r = 1.5 nm): ≈ 49 % of atoms on the surface
```

2. **Melting point.** Surface atoms are less tightly bound, so the melting point falls with size (gold nanoparticles of a few nm melt far below the bulk 1064 °C).

3. **Optical.** Gold nanoparticles look red, not gold: conduction electrons oscillate collectively with light (surface plasmon resonance, near 520 nm for ~20 nm spheres). Changing size or shape shifts the colour. Semiconductor nanocrystals change colour through quantum confinement.

4. **Electrical.** Confinement quantizes energy levels, so conductance changes in steps and the band gap grows as particles shrink.

5. **Magnetic.** Below ~20 nm, magnetic particles (Fe_3O_4) become single-domain and superparamagnetic: strongly magnetic in a field, but with no remanence when it is removed, so they do not clump.

6. **Mechanical.** Smaller grains in a metal block dislocation motion, so strength rises as grain size falls (Hall–Petch relation).

Types of nanomaterials

S11 · Lecture

By dimension

```
0D  all three dimensions nano:   nanoparticles, quantum dots, fullerene C60
1D  two dimensions nano:         nanotubes, nanowires, nanorods
2D  one dimension nano:          graphene, MoS2 sheets, thin films
3D  bulk with nano features:     nanocomposites, nanoporous materials
```

Fullerene C_{60}: 60 carbon atoms in a football shape (12 pentagons, 20 hexagons), about 0.7 nm across. Carbon nanotubes: graphene sheets rolled into cylinders (single- or multi-walled). The way the sheet is rolled, labelled by (n, m), decides the electrical behaviour: metallic if n − m is divisible by 3, otherwise semiconducting. They are extremely strong along their length and conduct heat well. Graphene: a single layer of carbon atoms in a hexagonal lattice, very strong, transparent and highly conductive, but with zero band gap.

By composition

Carbon-based (above); metals (Au, Ag, Pt); metal oxides (TiO_2, ZnO, Fe_3O_4); semiconductor nanocrystals (CdSe, InP); polymeric and lipid nanoparticles (liposomes, dendrimers); ceramic.

Quantum dots

S12 · Lecture

Quantum dots are semiconductor nanocrystals of about 2–10 nm. They are so small that the electron and hole created by light are squeezed into a space smaller than they would naturally occupy in bulk (the exciton Bohr radius). This is quantum confinement.

The particle-in-a-box connection

From Unit 1, a confined particle has energies proportional to $1/L^2$. Shrinking the crystal raises the confinement energy of both electron and hole, so the gap grows. The Brus equation expresses this:

```
E_g(R) ≈ E_g(bulk) + (ħ²π²/2R²)(1/mₑ + 1/mₕ) − 1.8 e²/(4πε₀εR)
                       ^ confinement, grows as 1/R²      ^ Coulomb attraction,
smaller correction
```

Result: one material, many colours. CdSe has a bulk gap of 1.74 eV (about 710 nm, red/infrared). Smaller dots emit green or blue, larger dots red, simply by changing size. For emission at 2.48 eV, λ = 1240/2.48 = 500 nm. Dots have narrow, tunable emission, high brightness and good stability.

Uses: QLED displays and televisions, bio-imaging labels, LEDs, solar cells and photodetectors. The 2023 Nobel Prize in Chemistry honoured their discovery and synthesis. Concern: Cd and Pb are toxic, so cadmium-free dots (InP) and carbon dots are being developed.

Applications of nanomaterials

S13 · Tutorial

1. **Medicine.** Liposomes and polymer particles carry drugs to targets; lipid nanoparticles deliver mRNA vaccines; iron-oxide particles enhance MRI; gold nanoshells heat and destroy tumour cells under near-infrared light.
2. **Electronics.** Transistors with nanometre features, flash memory, flexible conductive inks, nanowire sensors.
3. **Energy.** Nanostructured electrodes for Li-ion batteries, nanocatalysts in fuel cells, quantum dot and perovskite solar cells.
4. **Environment.** TiO_2 photocatalysts break down pollutants, nanomembranes filter water, nanoscale iron treats contaminated groundwater.
5. **Consumer and materials.** ZnO/TiO_2 transparent sunscreens, silver antibacterial fabrics, self-cleaning glass, nano-silica in tyres, stronger lighter composites.
6. **Risks.** Particles small enough to enter cells or be inhaled may behave differently from the bulk material, so toxicity and environmental fate must be tested, and release should be controlled.

Practical: strength of an acid with a pH meter

S4–S5 · Practice

A glass electrode's potential changes by 0.0592 V per pH unit (Unit 3), so a pH meter turns a voltage into pH after calibration.

1. **Calibrate.** Using pH 4.00, 7.00 and 10.00 buffers; rinse the electrode with distilled water between solutions.
2. **Titrate.** Put 25 mL of acid in a beaker, add 0.1 M NaOH from the burette in 1 mL steps (0.2 mL near the jump), stir, and record pH after each addition.
3. **Find the end point.** Plot pH vs V, and also ΔpH/ΔV vs V; the peak is the end point.
4. **Calculate.**

```
N₁V₁ = N₂V₂   (acid = NaOH)
Example: 25 mL HCl, 22.5 mL of 0.1 N NaOH: N = 0.1×22.5/25 = 0.090 N
Strength = N × equivalent mass = 0.090 × 36.46 = 3.28 g/L
```

Strong vs weak acid curves. A strong acid (HCl) starts at pH ≈ 1 and rises steeply through pH 7 at the end point. A weak acid (CH_3COOH) starts higher, has a flatter buffer region, and its end point is above 7 (about 8.7) because the acetate formed is basic. At the half-equivalence point [HA] = [A^-], so by the Henderson–Hasselbalch equation

```
pH = pKa + log([A⁻]/[HA])   →   pH(half-equivalence) = pKa
acetic acid: pKa = 4.76, Ka = 1.8×10⁻⁵
```

Practical: band gap of a semiconductor

S9–S10 · Practice

For an intrinsic semiconductor $R = R_0 \exp(E_g/2kT)$. Taking logs gives a straight line:

```
ln R = ln R₀ + (E_g/2k)(1/T)
```

1. **Measure.** Heat a semiconductor sample (Ge crystal or a thermistor) slowly in an oil bath; record its resistance and temperature at regular steps, with the thermometer touching the sample.
2. **Convert and plot.** T in kelvin; plot ln R against 1/T.
3. **Slope.** m = E_g/2k, so E_g = 2k·m. With k = 8.617×10^{-5} eV/K:

   ```
   E_g (eV) = 1.723×10⁻⁴ × m(K)
   Example: m = 3900 K → E_g = 0.67 eV (germanium)
   ```

4. **Check.** Use only the high-temperature (intrinsic) region; at low T, impurity levels dominate and the slope is smaller. Errors: lag between the thermometer and the sample, and self-heating from the measuring current.

An optical alternative (Tauc plot from UV–Vis absorption) appears in Unit 5.

Practical: HCl and acetic acid mixture by conductometry

S14–S15 · Practice

Conductance G = 1/R measures how well ions carry current. Each ion contributes in proportion to its concentration and its mobility. Ionic conductivity (S cm^2 mol^{-1}): H^+ 350, OH^- 199, Na^+ 50, Cl^- 76, CH_3COO^- 41. H^+ and OH^- are unusually mobile.

Titrating the mixture with NaOH

1. **Strong acid first.** HCl is neutralized: H^+ (350) is replaced by Na^+ (50), so conductance falls steeply.
2. **Then the weak acid.** After HCl is gone, acetic acid (barely ionized) turns into sodium acetate (a salt with Na^+ and acetate ions), so conductance rises slowly.
3. **Excess base.** Once both acids are neutralized, free OH^- (199) accumulates and conductance rises sharply.

Plot conductance (corrected for dilution) against NaOH volume; three straight lines meet at two break points. V_1 is the end of HCl, V_2 the end of acetic acid.

```
N(HCl) = N(NaOH) × V₁ / V(sample)
N(CH₃COOH) = N(NaOH) × (V₂ − V₁) / V(sample)
Example: 0.1 N NaOH, V₁ = 12.0 mL, V₂ = 20.0 mL, 25 mL sample:
  HCl = 0.1×12.0/25 = 0.048 N     acetic acid = 0.1×8.0/25 = 0.032 N
```

Practical tips: keep the temperature constant, stir, use a more concentrated titrant so the volume change is small, and rinse the cell. An advantage of this method is that it needs no indicator, so it works for coloured or turbid solutions, and the break points are found from the graph rather than a colour change.

Unit 5: Spectroscopic Analysis

How light and radio waves reveal the structure of a molecule: UV-Vis, IR and NMR.

Electromagnetic radiation

S1 · Lecture

Spectroscopy studies how matter absorbs, emits or scatters electromagnetic radiation. Radiation is an oscillating electric and magnetic field travelling at $c = 3.00\times10^{8}$ m/s. It is described by wavelength λ (distance between peaks), frequency ν (oscillations per second), and wavenumber $\tilde{\nu} = 1/\lambda$ (waves per cm, the unit used in IR). It is also made of photons, each of energy $E = h\nu$.

```
c = λν      E = hν = hc/λ      ν̃ = 1/λ      E(eV) ≈ 1240/λ(nm)
```

Example, green light of 500 nm: $\nu = 3.00\times10^{8} / 5.00\times10^{-7} = 6.0\times10^{14}$ Hz; $E = (6.626\times10^{-34})(6.0\times10^{14}) = 3.98\times10^{-19}$ J = 2.48 eV = 239 kJ/mol, comparable to the energy of a chemical bond. This is why light can drive chemistry.

Regions of the spectrum and what each does to a molecule

```
Radio waves    (60–900 MHz)     flip nuclear spins                    → NMR
Microwave                         change molecular rotation             → microwave
spectroscopy
Infrared       (4000–400 cm⁻¹)   change bond vibration                 → IR
Visible/UV     (700–200 nm)      excite valence electrons              → UV–Vis
X-ray          (< 10 nm)         excite or remove inner-shell electrons → XPS, X-ray
```

Higher frequency means higher energy. In an experiment, a spectrum is a plot of how much radiation is absorbed (absorbance) against wavelength, frequency or wavenumber. A molecule absorbs only when the photon energy matches an energy gap in the molecule exactly ($\Delta E = h\nu$), so the spectrum is a fingerprint of its energy levels. Transmittance $T = I/I_0$ and absorbance $A = -\log_{10}T$ are the two ways of reporting it.

Vibrational and electronic transitions

S2 · Lecture

A molecule's energy is quantized in three ways, with very different sizes:

```
E_total ≈ E_electronic + E_vibrational + E_rotational
electronic  ≈ 1–10 eV      (UV–Vis)
vibrational ≈ 0.05–0.5 eV (IR, 400–4000 cm⁻¹)
rotational  ≈ 0.001 eV    (microwave)
```

Vibrational levels

A bond behaves like a spring with levels E_v = (v + ½)hν, v = 0, 1, 2, … The lowest level has the zero-point energy ½hν. At room temperature kT = 0.0259 eV ≈ 208 cm^{-1}, so the fraction of C=O bonds (1700 cm^{-1}) in v = 1 is e^(−1700/208) ≈ 3×10^{-4}: almost every molecule starts in v = 0, and the main IR absorption is v = 0 → 1.

Electronic levels

Absorbing a UV or visible photon moves an electron from an occupied orbital (usually the HOMO) to an empty one (LUMO). Each electronic state has its own set of vibrational levels, so a UV band is broad: electronic change plus vibrational changes at once.

Selection rules

Not every energy match leads to absorption. IR: the vibration must change the molecule's dipole moment (HCl yes; N_2 and O_2 no). Electronic: spin must not change ($\Delta S = 0$), and transitions that are symmetry-forbidden (such as $n \rightarrow \pi^*$ in carbonyls) are weak. NMR: only nuclei with spin ≠ 0. Selection rules explain why some absorptions are strong, others weak, and others missing.

The Born–Oppenheimer approximation

S3 · Tutorial

The full Schrödinger equation for a molecule has electrons and nuclei all moving and interacting, which is impossible to solve exactly except for the simplest cases. The Born–Oppenheimer (BO) approximation simplifies it using one fact: a proton is about 1836 times heavier than an electron (and other nuclei are heavier still).

1. **Compare speeds.** Nuclei move slowly; electrons move so fast that they adjust immediately to any change in the nuclear positions. Compare a heavy truck with a swarm of bees around it.

2. **Freeze the nuclei.** Treat the nuclear positions R as fixed numbers and solve the electronic Schrödinger equation (electron kinetic energy, electron–nucleus attraction, electron–electron repulsion, plus the fixed nuclear repulsion):

   ```
   Ĥ_el ψ_el(r; R) = E_el(R) ψ_el(r; R)
   ```

3. **Repeat for many geometries.** Plotting E_el(R) against the nuclear separation gives a potential energy curve (surface for bigger molecules), with a minimum at the equilibrium bond length.

4. **Let the nuclei move on that curve.** E_el(R) acts as the potential energy in a second, nuclear Schrödinger equation, whose solutions are the vibrational and rotational levels.

5. **Result.**

   ```
   ψ_total ≈ ψ_el(r; R) × ψ_nuclear(R)
   E_total = E_el + E_vib + E_rot
   ```

Consequences. (1) Energy separates into electronic, vibrational and rotational parts, which is what lets spectroscopy treat each separately. (2) The concept of a molecule having a shape and a bond length exists because of the potential energy surface. (3) Computational chemistry (Unit 2 tools) calculates E_el(R) to optimize geometries.

The Franck–Condon principle

Absorption of a photon takes about 10^{-15} s, far shorter than a vibration (10^{-13} s). The nuclei have no time to move, so the transition is "vertical" on the potential energy diagram. The upper-state vibrational level reached is the one whose wavefunction overlaps best with the ground state, which creates the vibrational fine structure in UV spectra.

Where it fails

When two electronic surfaces come close or cross (photochemistry, conical intersections), electron and nuclear motion are coupled. It is also less accurate for very light nuclei (hydrogen tunnelling).

UV–visible spectroscopy

S6 · Lecture

Beer–Lambert law, derived

1. **Light is absorbed layer by layer.** A thin slice dl of solution of concentration c absorbs a fraction of the light proportional to c·dl:

   ```
   dI/I = −k c dl
   ```

2. **Integrate from 0 to l.**

   ```
   ln(I₀/I) = k c l   →   log₁₀(I₀/I) = (k/2.303) c l
   ```

3. **Name the terms.** A is absorbance, ε the molar absorptivity (L mol^{-1} cm^{-1}), c concentration (mol/L), l path length (cm).

   ```
   A = ε c l
   ```

Example: A = 0.45 at λmax for ε = 15 000, cell 1.00 cm: c = 0.45/(15 000 × 1) = 3.0×10^{-5} M. Transmittance T = 10^(−0.45) = 35.5 %. The law holds for dilute solutions (A below about 1) and a single absorbing species.

Electronic transitions in molecules

```
σ → σ*   < 200 nm     alkanes (far UV, not seen)
n → σ*   ~ 180–250    O, N, S lone pairs (alcohols, amines)
π → π*   170–250+     alkenes, aromatics; strong (ε 10³–10⁵); moves to longer λ as
conjugation grows
n → π*   270–300      C=O, N=O; weak (ε 10–100), symmetry-forbidden
```

Chromophore: the part of a molecule that absorbs (C=C, C=O, benzene ring). Auxochrome: a group (–OH, $–NH_2$) that shifts and intensifies absorption. Shifts: bathochromic (red, to longer λ), hypsochromic (blue), hyperchromic (more intense), hypochromic (less).

Why conjugation lowers the energy (particle-in-a-box check)

Model the π electrons of a polyene as N electrons in a 1-D box of length L (Unit 1). The HOMO is n = N/2 and LUMO n = N/2 + 1, so

```
ΔE = (N + 1) h² / 8mL²
Butadiene: N = 4, L ≈ 5.6 Å:
```

```
ΔE = 5(6.626×10⁻³⁴)² / [8(9.109×10⁻³¹)(5.6×10⁻¹⁰)²] = 9.6×10⁻¹⁹ J = 6.0 eV
λ = 1240 / 6.0 ≈ 207 nm     (observed 217 nm)
```

As the chain lengthens, L grows and N grows, but L^2 grows faster, so ΔE falls, and λ shifts to the red: ethylene 165 nm, butadiene 217, hexatriene 258, octatetraene 290, β-carotene (11 C=C) about 450 nm, so it absorbs blue and looks orange. The colour we see is the complement of the colour absorbed.

Solvent and instrument

Polar solvents stabilize the n-orbital, so n → π* shifts to shorter λ (blue), while π → π* usually moves slightly red. Instrument path: lamp (deuterium for UV, tungsten for visible) → monochromator → sample cuvette (quartz for UV) → detector. Always record a blank of the solvent first.

Infrared spectroscopy

S7 · Lecture

Hooke's law: vibrational frequency of a bond

1. **Model.** Two atoms joined by a spring of force constant k (N/m). The restoring force is F = −kx.
2. **Equation of motion.** For two masses, replace them by the reduced mass $\mu = m_1m_2/(m_1 + m_2)$. Newton's law $\mu\, d^2x/dt^2 = -kx$ has oscillating solutions with

   ```
   ν = (1/2π)√(k/μ)      ν̃= (1/2πc)√(k/μ)      ν̃(cm⁻¹) = 130.3 √(k(N/m) / μ(amu))
   ```

3. **Meaning.** Stiffer bonds (large k) and lighter atoms (small μ) vibrate faster.

```
HCl: k = 516 N/m, μ = (1×35)/36 = 0.972 amu → ν̃= 130.3√(516/0.972) = 3002 cm⁻¹
(observed fundamental 2886)
DCl: μ = 1.892 amu → 130.3√(516/1.892) = 2152 cm⁻¹ (isotope shift, ratio
√(0.972/1.892) = 0.72)
C–C 450, C=C 960, C≡C 1560 N/m with μ = 6 → 1128, 1648, 2101 cm⁻¹
```

Higher bond order means a higher frequency. Real bonds are slightly anharmonic: levels crowd together at high v and the bond can dissociate; this allows weak overtones (Δv = ±2) and makes the observed fundamental lower than the harmonic value.

Which vibrations appear?

Dipole must change. A non-linear molecule has 3N − 6 vibrational modes (H_2O: 3: symmetric stretch, asymmetric stretch, bend); a linear one has 3N − 5 (CO_2: 4). The symmetric stretch of CO_2 gives no dipole change and is IR-inactive, while its asymmetric stretch (2349 cm^{-1}) is strongly active, which is how CO_2 traps heat in the atmosphere.

Group frequencies (cm^{-1})

```
O–H alcohol          3200–3600 broad      O–H acid         2500–3300 very broad
N–H amine           3300–3500            C–H sp³          2850–3000
C–H sp² / aromatic  3000–3100            C≡C or C≡N       2100–2260
C=O ketone, aldehyde 1700–1730 strong    C=O ester        1735–1750
C=C                 1620–1680            C–O              1050–1300
Below 1500: fingerprint region, unique to each molecule
```

How to read a spectrum

1. **Look at 3600–2500.** Broad band = O–H; two spikes = NH_2; 3000 vs 2900 separates sp^2 from sp^3 C–H.
2. **Check 2300–2000** for triple bonds.
3. **Check 1800–1650** for a strong C=O. Position tells the type.
4. **Use the fingerprint region** to confirm by matching with a reference.

Examples: acetone shows one strong peak near 1715 cm^{-1}; ethanol shows a broad 3300 and a C–O near 1050; a carboxylic acid combines a very broad 2500–3300 with 1710. Practical: FTIR with a KBr pellet, a thin liquid film, or ATR (a crystal pressed against the sample, minimal preparation).

Band gap from UV–visible spectra (Tauc plot)

S8 · Tutorial

For semiconductors and nanoparticles (Unit 4), UV–Vis gives the band gap. Absorption begins when photon energy exceeds E_g, because an electron is promoted from VB to CB. The Tauc relation describes the absorption edge:

```
(α h ν)^(1/γ) = B (h ν − E_g)
γ = 1/2 for direct allowed gaps (plot (αhν)²),  γ = 2 for indirect gaps (plot (αhν)^½)
```

1. **Record the spectrum** of the sample (a thin film, or a dispersion with a solvent blank).
2. **Convert wavelength to energy.**

   ```
   hν (eV) = 1240 / λ (nm)
   ```

3. **Convert absorbance to absorption coefficient.** For a dispersion, A itself can be used for a qualitative edge.

   ```
   α = 2.303 A / t      (t = film thickness or cell path)
   ```

4. **Plot $(\alpha h\nu)^2$ against hν.** The curve rises steeply above the edge and has a straight section.
5. **Extrapolate the straight section to $(\alpha h\nu)^2 = 0$.** The intercept on the energy axis is E_g.

Example: ZnO nanoparticles with an absorption edge near 380 nm: E_g ≈ 1240/380 = 3.26 eV (bulk ZnO is about 3.3 eV, direct). Smaller particles show an edge at shorter wavelength, which shows the quantum-confinement blue shift of Unit 4.

Nuclear magnetic resonance: introduction

S11 · Lecture

Many nuclei have spin, so they behave as tiny magnets. ^{1}H, ^{13}C, ^{19}F and ^{31}P have spin I = ½ (two states, m = +½ and −½). ^{12}C and ^{16}O have I = 0 and are silent. Without a field the two states have equal energy; in a strong magnetic field B_0 they split:

```
ΔE = γħB₀ = hν      Larmor frequency ν = γB₀/2π
¹H: γ = 2.675×10⁸ rad s⁻¹ T⁻¹.   At B₀ = 7.05 T: ν = (2.675×10⁸)(7.05)/2π = 3.0×10⁸
Hz = 300 MHz
```

Resonance: a radio-frequency pulse with exactly this frequency flips nuclei from the lower (aligned) state to the upper. Magnets are labelled by their ^{1}H frequency (300, 400, 600 MHz). Population difference: $\Delta N/N \approx h\nu/2kT = (6.626\times10^{-34})(3\times10^{8})/(2 \times 1.381\times10^{-23} \times 298) \approx 2.4\times10^{-5}$, only about 24 per million, so NMR is a low-sensitivity technique and needs many scans or strong magnets.

How a modern spectrometer works

1. **Place the sample in a strong superconducting magnet.** (Unit 4: a Nb–Ti solenoid, cooled by liquid helium.)
2. **Send a short radio pulse** that excites all nuclei at once.
3. **Record the decaying signal** from the nuclei as they relax (the free-induction decay, FID).
4. **Fourier-transform the FID** into a spectrum of intensity against frequency.

Samples are dissolved in deuterated solvents ($CDCl_3$, D_2O) so that the solvent gives no ^{1}H signal. Tetramethylsilane (TMS) is the reference.

NMR explained: chemical shift, integration and splitting

S12 · Lecture

1. Chemical shift: what kind of environment?

Electrons around a nucleus circulate in the field and create a small opposing field, so the nucleus feels $B_{eff} = B_0(1 - \sigma)$, where σ is the shielding. Nuclei with more electron density are more shielded and resonate at lower frequency. To make the scale independent of magnet strength, shifts are reported relative to TMS:

```
δ (ppm) = (ν_sample − ν_TMS) / ν_spectrometer × 10⁶      TMS: δ = 0
```

Example: a peak 1500 Hz from TMS on a 300 MHz instrument has δ = 1500/300 = 5.0 ppm (on a 600 MHz one it would be 3000 Hz but still 5.0 ppm). Electronegative neighbours remove electrons (deshielding) and increase δ.

```
¹H:  R–CH₃ 0.9   –CH₂– 1.3   C=C–CH₃ 1.7   CH₃–C=O 2.1   N–CH 2.3–3   O–CH 3.3–4.5
     alkene C–H 4.6–6   aromatic 6.5–8   aldehyde 9–10   carboxylic acid 10–12
```

2. Integration: how many hydrogens?

The area under each signal is proportional to the number of hydrogens giving it. Ratios, such as 3 : 2 : 3, are read directly.

3. Splitting (spin–spin coupling): what are the neighbours?

A hydrogen feels the small magnetic fields of H atoms on adjacent carbons (three bonds away), which splits its signal into n + 1 lines when there are n equivalent neighbours. Intensities follow Pascal's triangle. The spacing is the coupling constant J (Hz), the same for the two coupled partners.

```
0 neighbours: singlet (s)   1: doublet 1:1   2: triplet 1:2:1   3: quartet 1:3:3:1
4: quintet
```

Equivalent hydrogens do not split each other, and O–H protons usually appear as a broad singlet because they exchange rapidly.

Examples

```
Ethanol CH₃CH₂OH:  CH₃  δ 1.2  triplet, 3H  (two CH₂ neighbours)
                   CH₂  δ 3.7  quartet, 2H  (three CH₃ neighbours)
                   OH   δ ~2.6 broad singlet, 1H
Ethyl acetate: CH₃C=O δ 2.0 s 3H;  OCH₂ δ 4.1 q 2H;  CH₃ δ 1.25 t 3H
```

^{13}C NMR

Only 1.1 % of carbon is ^{13}C, so the signals are weak, but the spectra are simple: with proton decoupling, every different carbon gives one line. The number of lines equals the number of distinct carbon environments (ethanol 2; benzene 1). Shifts: alkyl 10–50, C–O 50–90, alkene and aromatic 110–150, C=O 160–220 ppm.

Applications of NMR, with a worked problem

S13 · Tutorial

Solving a structure: $C_4H_8O_2$

1. **Degrees of unsaturation.** (2C + 2 − H)/2 = (10 − 8)/2 = 1, so there is one C=O or one ring.
2. **Read the ^{1}H spectrum.** Signals: δ 1.25 (triplet, 3H), δ 2.0 (singlet, 3H), δ 4.1 (quartet, 2H). Total 8H matches the formula.
3. **Assign.** A triplet 3H next to a quartet 2H is an ethyl group, CH_3–CH_2–. The quartet at 4.1 is far downfield, so the CH_2 is attached to oxygen: –O–CH_2CH_3. The 3H singlet at 2.0 has no neighbours and sits at the value for CH_3–C=O.
4. **Assemble.** $CH_3C(=O)$–O–CH_2CH_3, ethyl acetate. Confirm with IR (strong C=O near 1740 cm^{-1}, no O–H).

Wider uses

Structure determination in organic and natural-product chemistry; purity and quantification (peak area is directly proportional to the number of nuclei); reaction monitoring; pharmaceuticals and drug screening; protein and nucleic-acid structure with multi-dimensional NMR; metabolomics. MRI is NMR of water protons: magnetic field gradients label position, and relaxation times differ among tissues, giving images without ionizing radiation.

How the three techniques complement each other

```
UV–Vis: conjugation and concentration       IR: functional groups
NMR: complete carbon–hydrogen skeleton    (together they identify an unknown)
```

Practical: Hooke's-law IR spectra simulation

S4–S5 · Practice

Aim: predict the IR frequency of diatomic bonds from k and μ, and draw a simulated spectrum.

1. **Collect data.**

```
k (N/m):  HF 966   HCl 516   HBr 412   CO 1857   C–H ~500
```

2. **Compute μ in amu,** then $\tilde{\nu} = 130.3\sqrt{(k/\mu)}$. Examples: HF (μ = 0.957) gives 4140; CO (μ = 6.857) gives 2145 cm^{-1} (observed 2143).

3. **Isotope effect.** Replace H with D: μ almost doubles, so $\tilde{\nu}$ falls by about $1/\sqrt{2}$ (HCl 3002 → DCl 2152).

4. **Plot.** Draw each absorption as a Lorentzian peak:

```
import numpy as np, matplotlib.pyplot as plt
x = np.linspace(500, 4500, 4000)
peaks = [130.3*np.sqrt(k/mu) for k, mu in [(966,.957),(516,.972),(1857,6.857)]]
y = sum(1/(1+((x-p)/15)**2) for p in peaks)
plt.plot(x, y); plt.gca().invert_xaxis()   # IR spectra are plotted high→low
```

5. **Compare.** Harmonic values overestimate real frequencies by a few percent because of anharmonicity.

Practical: excitation energy of an organic molecule

S9–S10 · Practice

Aim: compute the first electronic excitation energy of a dye or conjugated molecule and compare with its colour.

1. **Build and optimize.** Draw the molecule (Avogadro, RDKit) and optimize the geometry with a quantum method (for example B3LYP/def2-SVP in ORCA or Psi4).

2. **Run a TD-DFT calculation** for the lowest 5–10 excited states.

3. **Read the output.** For each state: excitation energy E (eV), oscillator strength f (intensity; ignore f ≈ 0 states), and which orbitals are involved (usually HOMO → LUMO).

4. **Convert.**

```
λ (nm) = 1240 / E (eV)
```

5. **Compare with experiment** and with the colour seen: a molecule absorbing 450 nm (blue) appears yellow-orange.

6. **Study a trend.** Do ethylene → butadiene → hexatriene → octatetraene and watch λ increase (165, 217, 258, 290 nm), as the box model predicts.

Note: the HOMO–LUMO gap is only a rough guide; true excitation energies also include the attraction between the excited electron and the hole it leaves, so they differ from orbital-energy gaps. Calculated λ often differs from experiment by 10–30 nm.

Viva: questions and answers

S14–S15 · Practice

1. **Why must photon energy equal the energy gap?** Energy levels are quantized, so only $\Delta E = h\nu$ is absorbed.
2. **Which transition does IR cause, and which UV–Vis?** IR: vibrational (v = 0 → 1). UV–Vis: electronic (HOMO → LUMO).
3. **Why is N_2 IR-inactive?** Its stretching vibration does not change the dipole moment (zero before and after).
4. **What does the Born–Oppenheimer approximation assume?** Nuclei are far heavier and slower, so electronic and nuclear motion can be separated.
5. **State the Beer–Lambert law.** $A = \varepsilon cl$: absorbance is proportional to concentration and path length.
6. **Why do longer conjugated systems absorb at longer wavelength?** The HOMO–LUMO gap shrinks (box length grows).
7. **In Hooke's law, what happens to $\tilde{\nu}$ when the bond is stiffer, or atoms heavier?** Stiffer: increases. Heavier: decreases (μ in the denominator).
8. **Why use TMS in NMR?** Single sharp signal at δ 0, inert, volatile, and its protons are highly shielded.
9. **What is the n + 1 rule?** A signal coupled to n equivalent neighbouring H splits into n + 1 lines.
10. **Why is NMR less sensitive than UV–Vis?** The energy gap is tiny, so the population difference between spin states is only ~ 10^{-5}.
11. **How do you find a band gap from UV–Vis?** Tauc plot: $(\alpha h\nu)^2$ against $h\nu$; extrapolate the straight part to the energy axis.
12. **How does an ethyl group appear in ^{1}H NMR?** A triplet (3H) and a quartet (2H).